

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

VII SEMESTER

R22

V SEMESTER		REMARKS				
Course Code	Title of the Course	L	T	P/D	CH	C
22PC1CY401	Blockchain Technology	3	0	0	3	3
22PC1CY402	Security Operations Center	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1CY401	Cyber Threat Intelligence	3	0	0	3	3
22PC1AM303	Neural Networks and Deep Learning					
22PE1DS402	Distributed Computing					
22PE1AM403	Information Retrieval Systems					
22PE1DS401	Natural Language Processing and Text Analytics					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Cyber Security Stream	3	0	0	3	3
	Artificial Intelligence Stream					
	Machine Learning Stream					
	Programming Stream					
	Artificial Intelligence Stream					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2CY401	Blockchain Technology Laboratory	0	0	2	2	1
22PC2CY402	Security Operation Center Laboratory	0	0	2	2	1
22PW4CY401	Major Project Phase – I	0	0	6	6	3
Total		15	0	10	25	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
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VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1CY402	Web and Database Security	3	0	0	3	3
22PE1AM405	Cognitive Engineering					
22PE1CS404	Augmented Reality and Virtual Reality					
22PE1IT402	Software Project Management					
22PE1AM402	Reinforcement Learning					
	PROFESSIONAL ELECTIVE – VI					
22PE1CY403	Security in Internet of Things	3	0	0	3	3
22PE1IN303	Adhoc and Sensor networks					
22PE1DS406	Quantum Computing					
22PE1AM407	Robotic Process Automation					
22PE1CS405	Social Media Analytics					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4CY402	Major Project Phase – II	0	0	22	22	11
Total		9	0	22	31	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1CY401) BLOCKCHAIN TECHNOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To provide an understanding of blockchain technology, including its origin, components, types, consensus mechanisms, and cryptocurrency applications
- To explore public blockchain systems and smart contracts, their principles, implementations, and industry applications
- To know the security issues of blockchain technology
- To understand security challenges in blockchain and explore industry applications.
- To study practical blockchain implementations through case studies

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze blockchain technology by examining its origin, components, types, consensus mechanisms, and cryptocurrency applications

CO-2: Evaluate public blockchains and smart contracts, their principles, implementations, and industry applications

CO-3: Examine private and consortium blockchains, their applications, and the role of ICOs in blockchain fundraising

CO-4: Apply blockchain security principles and explore its applications, limitations, and future scope

CO-5: Implement blockchain solutions through case studies and development with Python and Hyperledger Fabric

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	1	1	1	2	1	1	1	2	2	1	1
CO-2	3	2	3	3	3	2	2	3	2	1	3	3	1	3	3
CO-3	2	2	2	1	1	1	1	2	1	1	1	2	1	2	1
CO-4	3	3	2	1	1	1	1	2	1	1	2	3	1	2	2
CO-5	3	3	3	1	1	1	1	1	1	1	1	2	1	2	1

UNIT-I:

Fundamentals of Blockchain: Introduction, Origin of Blockchain, Blockchain Solution, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency – Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.

UNIT-II:

Public Blockchain System: Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts: Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.

UNIT-III:

Private Blockchain System: Introduction, Key Characteristics of Private Blockchain, Why We Need Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain.

Consortium Blockchain: Introduction, Key Characteristics of Consortium Blockchain, Why We Need Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda. Initial Coin Offering: Introduction, Blockchain Fundraising Methods, Launching an ICO, Investing in an ICO, Pros and Cons of Initial Coin Offering, Successful Initial Coin Offerings, Evolution of ICO, ICO Platforms.

UNIT-IV:

Security in Blockchain: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric.

Applications of Blockchain: Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.

UNIT-V:

Blockchain Case Studies: Case Study 1 – Retail, Case Study 2 – Banking and Financial Services, Case Study 3 – Healthcare, Case Study 4 – Energy and Utilities.

Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK.

TEXT BOOK:

1. Block Chain Technology, Chandramouli Subramanian, Asha A. George, Abhilash K. A. and Meena Karthikeyan, Universities Press

REFERENCES:

1. Blockchain Blueprint for Economy, Melanie Swan, SPD O'Reilly
2. Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gaur, Pearson Addison Wesley

ONLINE RESOURCES:

1. The Bitcoin Backbone Protocol - Analysis and Applications, J. A. Garay et al., EUROCRYPT2015LNCSVOI9057, Vol. II, pp. 281310 (eprint.iacr.org/2016/1048)

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1CY402) SECURITY OPERATIONS CENTER (SOC)

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the operational, tactical, and strategic components of a Security Operations Center (SOC)
- To learn tools, technologies, and methodologies for monitoring and defending against cyber threats
- To develop practical skills for analyzing security incidents and implementing mitigation measures
- To analyze SIEM, SOAR, and automation for threat detection and incident response in SOC
- To understand collaboration workflows within a SOC and regulatory compliance requirements

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze SOC fundamentals, including its structure, roles, components, workflows, and lifecycle

CO-2: Identify and evaluate threats using SOC tools and threat intelligence frameworks

CO-3: Analyze security incidents and recommend appropriate response actions

CO-4: Design a strategic SOC plan incorporating best practices and modern tools

CO-5: Implement SOC metrics to assess performance and compliance with regulations

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	3	2	1	2	3	1	3	2	1	1	3	3	3	1
CO-2	2	2	3	2	3	3	1	3	1	1	2	3	3	3	2
CO-3	1	2	2	1	2	3	1	3	3	2	1	3	3	3	1
CO-4	3	3	3	3	3	3	1	3	1	1	1	3	3	2	1
CO-5	2	2	2	2	2	2	1	3	1	1	1	3	3	3	3

UNIT-I:

Introduction to Security Operations Center (SOC): Definition, purpose, and importance of SOC in cybersecurity operations. SOC structure: Tiered operations (Tier 1, 2, 3), SOC team roles (Analysts, Engineers, Managers), SOC infrastructure: Physical and virtual components, network architecture, data storage, Key components: SIEM, IDS/IPS, firewalls, and other monitoring tools, SOC Lifecycle: Incident detection, analysis, response, and resolution. SOC workflows and processes: Incident detection, triage, analysis, escalation, response, and reporting.

UNIT-II:

Threat Intelligence and Detection Methods: Types of threat intelligence: Tactical, Operational, Strategic, integration into SOC workflows, Tools and resources for threat intelligence (MISP, OpenDXL, STIX/TAXII), Common cyber threats and attack vectors (Phishing, DDoS, Malware, Ransomware, etc.), Recognizing indicators of compromise (IoCs) and tactics, techniques, and procedures (TTPs) used by adversaries, Signature-based detection vs. Anomaly-based detection.

UNIT-III:

Incident Response: Phases of Incident Response: Preparation, Detection, Containment, Eradication, Recovery, and Documentation. SOC's role in incident management, Legal and ethical considerations in incident response, Communication protocols within a SOC during incidents (internal and external communication), Application Security in incident Response.

UNIT-IV:

Security Tools and Automation in SOC: Overview of SIEM platforms (e.g., Splunk, IBM QRadar, Elastic Stack), SIEM architecture: Data collection, normalization, correlation, and reporting, Log aggregation and analysis for threat detection, Introduction to Security Orchestration, Automation, and Response (SOAR), Benefits of automation in SOC (incident handling, data enrichment, ticketing systems), Automation tools (e.g., Phantom, Demisto, Cortex XSOAR), Techniques for proactive threat hunting to identify advanced persistent threats (APTs).

UNIT-V:

SOC Metrics, Reporting, and Compliance: Key performance indicators (KPIs) for SOC performance measurement: Mean time to detect (MTTD), Mean time to respond (MTTR), SOC maturity models (e.g., CMMI for SOC), Incident reporting: Creating reports, severity classification, and response time documentation, Reporting for stakeholders: Board-level reporting, regulatory compliance reporting, Compliance requirements: GDPR, HIPAA, PCI-DSS, DPDP Act-2023. SOC's role in ensuring compliance through monitoring and reporting.

TEXT BOOKS:

1. The Security Operations Center: Building, Operating, and Maintaining Your SOC, David Nathans
2. Network Security Through Data Analysis, Michael Collins

REFERENCES:

1. Applied Network Security Monitoring: Collection, Detection, and Analysis, Chris Sanders and Jason Smith
2. The Practice of Network Security Monitoring: Understanding Incident Detection and Response, Richard Bejtlich
3. Blue Team Handbook: Incident Response Edition, Don Murdoch

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CY401) CYBER THREAT INTELLIGENCE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basic concepts of cyber threat intelligence
- To examine various ways of data collection and intelligence reports
- To analyze the constant change of the threat environment as capabilities, intent, opportunities
- To gain knowledge on Threat hunting environment
- To apply MITRE attack framework to understand cyber adversary tactics and techniques

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the fundamental concepts of threat intelligence

CO-2: Analyze various forms of intelligence reports

CO-3: Understand different business models of threat actors, and how these dictate the choice of victims and the nature of their attacks

CO-4: Carry out atomic hunts to start the threat hunting process and understand the environment

CO-5: Perform advanced hunting using MITRE attack emulations

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	1	1	1	2	1	1	1	2	3	1	1
CO-2	3	2	3	3	3	2	2	3	2	1	3	3	3	1	3
CO-3	2	2	2	1	1	1	1	2	1	1	1	2	1	3	1
CO-4	3	3	2	1	1	1	1	2	1	1	2	3	1	3	3
CO-5	3	3	3	1	1	1	1	1	1	1	1	2	1	3	1

UNIT-I:

Introduction: Definition, Cyber Threat Intelligence Types, The intelligence cycle: Planning and targeting, Preparation and collection. Processing and exploitation, Analysis and production, Dissemination and integration, Evaluation and feedback.

Data Gathering and Collection: Indicators of compromise, Understanding malware, Using public sources for collection – OSINT, Honeypots, Malware analysis and sandboxing

UNIT-II:

Intelligence Collection: Hierarchy of Evidence, understanding Intelligence, third party Intelligence Reports, Internal Incident Reports: Creditability, Confidence, Understanding Errors. Root Cause Analysis, Active Intelligence Gathering.

UNIT-III:

Threat Environment: Threat Classification, Risk and Vulnerability, Threat Actors, Tactics, Techniques and procedures, Victimology: Diamond Model, Threat Landscape, Attack Vectors, Vulnerabilities, and Exploits: E-mail Attack Vectors, Web-based Attacks, Network Service Attacks, Supply Chain Attacks, The Kill Chain.

UNIT-IV:

Threat Hunting: Types of Threat Hunts, The Threat Hunter Skill Set, The Pyramid of pain, The Threat hunting Maturity Model, Threat Hunting Process: The Threat Hunting Loop, Threat hunting model, The data-driven methodology, Targeted Hunting Integrating Threat Intelligence

UNIT-V

MITRE Attack:

Framework: Tactics, techniques, sub-techniques, and procedures, The ATTACK Matrix, The ATTACK Navigator. MITRE CAR, MITRE Emulation plan.

TEXT BOOKS:

1. Martin Lee, Cyber Threat Intelligence, Wiley Publisher, 2023
2. Valentina Costa-Gazcón, Practical Threat Intelligence and Data-Driven Threat Hunting, Packt Publishing, 2021

REFERENCES:

1. Chad Maurice, Jeremy Thompson, William Copeland, The Foundations of Threat Hunting: Organize and Design Effective Cyber Threat Hunts to Meet Business Needs, 2022
2. Cyber Threat Intelligence: A Complete Guide, Gerardus Blokdyk, 2021
3. The Diamond Model of Intrusion Analysis, Threat Connect, Caltagirone S., Pendergast A., Betz C., 298 (0704), 1-61, 2013
4. MITRE ATTACK®: Design and Philosophy, Blake E. Strom, Andy Applebaum, Doug P. Miller, Kathryn C., Nickels Adam G., Pennington Cody, B. Thomas, The MITRE Corporation, 2020

ONLINE RESOURCES:

1. The Cycle of Cyber Threat Intelligence

B.Tech. VII Semester

(22PC1AM303) NEURAL NETWORKS AND DEEP LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand basic concepts of artificial neuron and its working principle
- To examine various types of feed-forward and feedback neural networks
- To understand basic functionality of deep neural networks
- To analyze various parameter tuning and optimization methods
- To apply various case studies of deep learning techniques

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand basic concepts of neural network architectures and learning rules

CO-2: Analyze feed-forward and feedback neural networks and its applications

CO-3: Identify various parameters and optimization strategies of deep learning models

CO-4: Analyze various metrics to optimize and fine tune deep learning models

CO-5: Apply various deep learning techniques to solve real-time applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	3	2	2	1	-	-	1	1	2	3	1	1
CO-2	2	3	3	3	2	1	1	-	2	-	1	1	3	1	-
CO-3	3	3	3	3	2	2	-	-	2	2	1	1	3	2	1
CO-4	2	3	2	2	-	1	-	-	-	-	1	1	3	1	1
CO-5	3	2	2	3	3	2	2	3	-	1	1	1	3	3	1

UNIT-I:

Introduction: Biological Neuron, Idea of computational units, McCulloch–Pitts unit and Thresholding logic, Linear Perceptron, Perceptron Learning Algorithm, Linear separability, Convergence theorem for Perceptron Learning Algorithm, Type of network architecture, Activation functions, Basic Learning rules.

UNIT-II:

Neural Networks: Basics, Functions in Neural Networks, Feedforward Neural Networks: Multilayer Neural Networks, Gradient Descent learning, Backpropagation, Empirical Risk Minimization, regularization, Radial Basis Neural Network.

Feedback Neural Networks: Introduction, Analysis of Linear Auto associative FF Networks, Analysis of Pattern Storage Networks, Stochastic Networks, and Simulated Annealing, Boltzmann Machine.

UNIT-III:

Deep Learning: Introduction to Deep Learning, Need of Deep Learning, Constraints of Deep Learning, Applications of Deep Learning and Classification of Deep Learning Architectures.

Convolution Neural Networks: Foundations of Convolutional Neural Networks – CNN operations – Architecture – Simple Convolution Network – Deep Convolutional Models – ResNet, AlexNet, InceptionNet, and others.

UNIT-IV:

Improving Deep Neural Networks: Data Augmentation - Under-fitting vs. over-fitting. Training Aspects of CNNs, Regularization, Weight Initialization, Activation Functions, Normalization, Hyperparameters in CNNs, Transfer Learning, and Fine Tuning in CNNs

UNIT-V:

Recurrent Neural Networks: Bidirectional RNNs, Long-Term Dependencies - Echo State Networks - Long Short-Term Memory and Other Gated RNNs - Optimization for Long-Term Dependencies - Explicit Memory.

Applications and Recent Trends: Attention in visual Models: Attention Networks and Transformer Networks, Deep Generative Models: VAEs, GAN

TEXT BOOKS:

1. Neural Networks and Deep Learning, Michael Nielsen, Determination Press
2. Deep Learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press, 2017

REFERENCES:

1. Deep Learning Step by Step with Python, N. D. Lewis, 2016
2. Deep Learning: A Practitioner's Approach, Josh Patterson, Adam Gibson, O'Reilly Media, 2017
3. Applied Deep Learning. A Case-based Approach to Understanding Deep Neural Networks, Umberto Michelucci, Apress, 2018
4. Deep Learning with TensorFlow: Explore Neural Networks with Python, Giancarlo Zacccone, Md. Rezaul Karim, Ahmed Menshawy, Packt Publishing, 2017

B.Tech. VII Semester

(22PE1DS402) DISTRIBUTED COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce concepts related to distributed computing systems
- To focus on performance and flexibility issues related to systems design decisions
- To expose students to current literature in distributed systems
- To prepare students for a real-world distributed application design

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand models of distributed computing

CO-2: Analyze algorithms for coordination, communication and synchronization in distributed systems

CO-3: Analyze distributed shared memory models

CO-4: Design and implement distributed file systems

CO-5: Design distributed algorithms for handling deadlocks

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	1	-	1	1	1	-	1	-	1	1	3	1
CO-2	2	2	1	1	1	1	1	1	-	-	-	-	1	2	1
CO-3	3	3	2	2	2	1	1	1	-	-	-	-	1	2	1
CO-4	1	1	3	3	2	1	1	1	-	-	1	1	1	1	-
CO-5	1	-	1	1	2	2	1	1	-	1	-	2	1	1	1

UNIT-I:

Distributed Computing Introduction: Definition, Relation to parallel systems, synchronous vs asynchronous execution, design issues and challenges .A Model of Distributed Computations: A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Logical Time: Logical clocks, scalar time, vector time, Matrix time, virtual time, Physical clock synchronization – NTP.

UNIT-II:

Global State and Snapshot Recording Algorithms: System model, Snapshot algorithms for FIFO channels, Variations of Chandy-Lamport algorithm, Snapshot algorithms for non-FIFO channels, Snapshots in a causal delivery system, Monitoring global state, Necessary and sufficient conditions for consistent global snapshots, Finding consistent global snapshots in a distributed computation.

Message Ordering and Group Communication: Message ordering paradigms, Group communication, Causal order (CO), Total order, Multicast, Propagation trees for

multicast, Application-level multicast algorithms, Fault-tolerant group communication, Multicast algorithms at the network layer.

UNIT-III:

Termination Detection: System model of a distributed computation, Termination detection using distributed snapshots, weight throwing and spanning-tree-based algorithms, Message-optimal termination detection, and Termination detection in a general distributed computing model, Termination detection in the atomic computation model, Termination detection in a faulty distributed system.

Distributed Mutual Exclusion Algorithms: Lamport's algorithm, Ricart–Agrawala algorithm, Singhal's dynamic information-structure algorithm, Lodha and Kshemkalyani's fair mutual exclusion algorithm, Quorum-based mutual exclusion algorithms, Maekawa's algorithm, Agarwal–El Abbadi quorum-based algorithm, Token-based algorithms, Suzuki–Kasami's broadcast algorithm, Raymond's tree-based algorithm.

UNIT-IV:

Deadlock Detection in Distributed Systems: System model, Models of deadlocks, Knapp's classification of distributed deadlock detection algorithms, Mitchell and Merritt's algorithm for the single resource model, Chandy–Misra–Haas algorithm for the AND model, Chandy–Misra–Haas algorithm for the OR model, Kshemkalyani–Singhal algorithm for the P-out-of-Q model .

Distributed Shared Memory: Abstraction and advantages, Memory consistency models, Shared memory mutual exclusion, Wait-freedom, Register hierarchy and wait-free simulations, Wait-free atomic snapshots of shared objects

UNIT-V:

Check Pointing and Rollback Recovery: Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, Koo–Toueg coordinated check pointing algorithm, Juang–Venkatesan algorithm for asynchronous check pointing and recovery, Manivannan–Singhal quasi-synchronous check pointing algorithm, Peterson–Kearns algorithm based on vector time, Helary–Mostefaoui–Netzer–Raynal communication-induced protocol.

TEXT BOOKS:

1. Distributed Computing, Ajay D. Kshemakalyani, Mukesh Singhal, Cambridge University Press, 2008
2. Distributed Systems - Principles and Paradigms, Andrew S. Tanenbaum, Maarten Van Steen, Prentice Hall of India, 2004

REFERENCES:

1. Distributed Computing: Fundamentals, Simulations, and Advanced Topics, Hagit Attiya, Jennifer Welch, Wiley
2. Distributed Systems: Principles and Paradigms, Andrew S. Tanenbaum, Maarten van Steen, Prentice Hall of India

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1AM403) INFORMATION RETRIEVAL SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To describe the differences between repositories like Database Management Systems, Information retrieval systems and data warehouse
- To discover various pre-processing techniques which can apply on text documents to outline the structure of queries and documents
- To articulate fundamental functions used in information retrieval such as automatic indexing, abstracting, clustering and visualizing data
- To learn the important concepts, algorithms, and data/file structures that are necessary to specify, design, and implement Information Retrieval (IR) systems
- To explore various text search algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify and understand the relationships between various Repository Systems

CO-2: Apply knowledge of data structures and indexing methods in information retrieval Systems

CO-3: Implement various clustering, searching techniques and information visualization techniques

CO-4: Analyze clustering techniques and algorithms using evaluation measures

CO-5: Implement various text search algorithms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	2	3	3	2	1	1	2	2	2	3	-	2
CO-2	3	3	3	3	3	2	2	2	2	2	2	2	2	3	2
CO-3	3	3	3	3	3	3	2	2	2	3	3	2	-	2	-
CO-4	3	3	3	3	3	3	2	2	2	3	3	2	3	-	-
CO-5	2	2	2	2	3	3	2	1	1	2	2	2	-	-	-

UNIT-I:

Introduction: Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses, Information Retrieval System Capabilities – Search, Browse, Miscellaneous.

UNIT-II:

Cataloging and Indexing: Objectives, Indexing Process, Automatic Indexing, Information Extraction

Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N – gram data structure, PAT data structure, Signature file structure, Hypertext data structure.

UNIT-III:

Automatic Indexing: Classes of automatic indexing, Statistical Indexing, Natural language, Concept indexing, Hypertext linkages.

Document and Term Clustering: Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters.

UNIT-IV:

User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, weighted searches of Boolean Systems, Searching the Internet and hypertext.

UNIT-V:

Text Search Algorithms: Introduction, Software text search algorithms, Hardware text search systems.

Information System Evaluation: Introduction, Measures used in system evaluation, Measurement example – TREC results.

TEXT BOOKS:

1. Information Storage and Retrieval Systems: Theory and Implementation, Kowalski, Gerald, Mark T. Maybury, Springer
2. Modern Information Retrieval: Ricardo Baeza, Yates, Pearson Education, 2007

REFERENCES:

1. Information Retrieval: Algorithms and Heuristics, David A. Grossman and Ophir Frieder, 2nd Edition, Springer
2. Information Retrieval Data Structures and Algorithms, Frakes W. B., Ricardo Baeza-Yates, Prentice Hall, 1992

B.Tech. VII Semester

(22PE1DS401) NATURAL LANGUAGE PROCESSING AND TEXT ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To create structured data out of free text content
- To implement recurrent network for language models
- To implement a PCA and ML algorithms in real life problems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Extract information from text automatically using concepts and methods from natural language processing (NLP) including stemming, n-grams, POS tagging, and parsing

CO-2: Analyze the text syntactically and semantically

CO-3: Develop speech-based applications that use speech analysis (phonetics, speech recognition, and synthesis)

CO-4: Analyze large volume text data generated from a range of real-world applications

CO- 5: Analyze EDA, feature extraction, machine learning models, visualization, and text processing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	2	1	2	2	2	3	-	-	-
CO-2	3	3	3	2	3	2	2	1	1	2	2	3	1	-	-
CO-3	3	2	2	2	3	1	1	1	3	2	2	2	-	-	-
CO-4	3	2	3	3	3	2	2	1	2	2	2	3	1	-	-
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	-	-	-

UNIT-I:

Data set overview Amazon Fine Food reviews (EDA), Data Cleaning Deduplication, Why convert text to vector, Bag of words(BoW),Text Preprocessing Stemming, Stop-word removal, Tokenization, Lemmatization, uni-gram, bi-gram, n-grams.

UNIT-II:

TF-IDF, Why use log in IDF, Word2Vec, Avg-Word2Vec, TF-IDF weighted Word2Vec, Bag of Words (Code Sample), text processing (Code Sample), Bi-Grams and n-grams (Code Sample).

UNIT-III:

TF-IDF (Code Sample), Avg-Word2Vec and TF-IDF-Word2Vec (Code Sample), Word2Vec, Word2Vec (Code Sample), Avg-Word2Vec and TFIDF-Word2Vec (Code Sample).

UNIT-IV:

Case Study: Quora question pair similarity problem, How to optimally learn from case-studies, Business Real world problem definition, Business objectives and constraints, Mapping to a data science problem and data overview, Mapping to a data science problem and performance metric.

UNIT-V:

Train-test split, EDA Basic Statistics, EDA Basic Feature Extraction, EDA Text processing, EDA Advanced Feature Extraction Feature analysis Data Visualization by T-SNE, EDA TF-IDF weighted Word2Vec futurization.

Data science Models Loading Data, data science Models Random Model, data science Models Logistic Regression and Linear SVM, data science Models XGBoost.

TEXT BOOKS:

1. Deep Learning in Natural Language Processing, Li Deng, Yang Liu, Springer
2. Introduction to Natural Language Processing, Jacob Eisenstein, MIT Press

REFERENCES:

1. Transformers for Natural Language Processing, Denis Rothman, Packt Publishing
2. Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, 1st Edition, O'Reilly Media
3. Deep Learning for NLP and Speech Recognition, Uday Kamath, John Liu, James Whitaker, 1st Edition, Springer
4. Handbook of Natural Language Processing, Nitin Indurkha, Fred J. Damerau, 2nd Edition, Chapman and Hall/CRC
5. Speech and Language Processing, Daniel Jurafsky, James Martin, 2nd Edition, Press Prentice Hall

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2CY401) BLOCKCHAIN TECHNOLOGY LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Basics of JavaScript /Java for Hyperledger Fabric, Basics of Solidity for ETH

COURSE OBJECTIVES:

- To understand blockchain fundamentals, distributed ledgers, consensus, smart contracts, and cryptography
- To apply ethereum development, solidity programming and smart contract deployment
- To explore hyperledger fabric, design, deploy, and manage permissioned blockchain networks
- To implement smart contracts, automate processes and execute agreements on the blockchain
- To analyze blockchain security and privacy, protect data and prevent attacks

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Configure Metamask to create wallets, manage accounts, and perform transactions on a test network

CO-2: Setup Ganache and integrate it with Metamask to create a custom RPC network and transfer ether

CO-3: Develop Solidity smart contracts for balance transfers and exception handling

CO-4: Deploy a Hyperledger Fabric network with multiple organizations and create communication channels

CO-5: Implement SDK programs to interact with deployed smart contracts and query blockchain data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	1	1	1	2	1	1	1	2	1	1	1
CO-2	3	2	3	3	3	2	2	3	2	1	3	3	2	1	2
CO-3	2	2	2	1	1	1	1	2	1	1	1	2	1	2	1
CO-4	3	3	2	1	1	1	1	2	1	1	2	3	1	2	1
CO-5	3	3	3	1	1	1	1	1	1	1	1	2	1	2	1

LIST OF PROGRAMS / EXERCISES:

WEEK 1 and 2:

- 1) Setup Metamask in the System and Create a wallet in the Metamask with Test Network.

WEEK 3:

- 2) Create multiple accounts in Metamask and perform the balance transfer between the accounts and describe the transaction specifications.

WEEK 4:

- 3) Setup the Ganache Tool in the system.

WEEK 5 and 6:

- 4) Create a custom RPC network in Metamask and connect it with Ganache tool and transfer the ether between ganache accounts.

WEEK 7:

- 5) Write a smart contract using a solidity program to perform the balance transfer from contract to other accounts.

WEEK 8:

- 6) Write a solidity program to perform the exception handling.

WEEK 9:

- 7) Setup the Hyperledger Fabric Network with 2 Organizations 1 Peer Each in the system.

WEEK 9:

- 8) Create a channel called mychannel, carchannel in the deployed network.

WEEK 10 and 11:

- 9) Take the existing Fabcar smart contract and add a new function to query the car on the basis of person name and deploy the smart contract on the Hyperledger Fabric Network.

WEEK 12 and 13:

- 10) Write an SDK program to query the person details from the deployed smart contract.

TEXT BOOKS:

1. Blockchain Blueprint for Economy, Melanie Swan
2. Blockchain Technology, Chandramouli Subramanian, Asha A. George, Abhilash K. A. and Meena Karthikeyan, Universities Press

REFERENCES:

1. Blockchain Basics: A Non-Technical Introduction in 25 Steps, Daniel Drescher, 1st Edition
2. Blockchain Blueprint for Economy, Melanie Swan, SPD O'Reilly
3. Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gaur, Pearson Addition Wesley

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2CY402) SECURITY OPERATIONS CENTER LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To understand the operational, tactical, and strategic components of a Security Operations Center (SOC)
- To learn tools, technologies, and methodologies for monitoring and defending against cyber threats
- To develop practical skills for analyzing security incidents and implementing mitigation measures
- To understand collaboration workflows within a SOC and regulatory compliance requirements
- To design and evaluate advanced detection strategies and incident response workflows

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Examine the roles, responsibilities, and workflows in a SOC

CO-2: Identify and evaluate threats using SOC tools and threat intelligence frameworks

CO-3: Analyze security incidents and recommend appropriate response actions

CO-4: Implement SOC metrics to assess performance and compliance with regulations

CO-5: Design a strategic SOC plan incorporating best practices and modern tools

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	3	2	1	2	3	1	3	2	1	1	3	3	3	-
CO-2	2	2	3	2	3	3	1	3	1	1	2	3	3	3	2
CO-3	1	2	2	1	2	3	1	3	3	2	1	3	3	3	3
CO-4	3	3	3	3	3	3	1	3	1	1	1	3	3	3	1
CO-5	2	2	2	2	2	2	1	3	1	1	1	3	2	3	-

LIST OF EXPERIMENTS:

- SIEM Installation and Setup-Install and configure a Security Information and Event Management(SIEM) such as ELK stack to centralize log collection and monitoring.
- Firewall/IDS/IPS Configuration-Configure and deploy security devices like Snort (IDS) and Pfsense(Firewall) for network defense.
- Writing SIEM Rules for Detection>Create detection rules using formats such as Sigma, SPL (Splunk Processing Language), KQL (Kibana Query Language), or DQL (Detection Query Language).

- 4) Log Collection and Parsing-Gather logs from various sources (Windows, IDS, firewall), parse them, and standardize formats using tools like Logstash, or Fluentd.
- 5) Detection of Suspicious Connections-Analyze network traffic to detect anomalous or malicious IP connections using Zeek.
- 6) Log Analysis and Anomaly Detection-Perform log analysis to identify abnormal activities or anomalies using ELK's Kibana dashboards.
- 7) Monitoring System Processes-Track and analyze system activities on Windows (Event Viewer) and Linux (auditd) to detect suspicious behaviors using SIEM.
- 8) SIEM Visualizations/Dashboards for Monitoring-Build and customize visualizations for real-time monitoring of security events in dashboards using Kibana.
- 9) Threat Intelligence for Analysis-Leverage threat intelligence platforms (Talos, X-Force, AlienVault OTX, or VirusTotal) to enrich log data with external threat indicators.
- 10) Phishing Analysis-Examine phishing emails and malicious URLs using tools like PhishTank, URLScan, Mxtoolbox and mha.azurewebsites.net
- 11) Brute Force Detection-Identify repeated login failures in logs to detect brute force attempts, using SIEM alerts.
- 12) SOC Reports and Documentation-Generate actionable SOC reports, including summaries of incidents, detection metrics, and operational recommendations.

TEXT BOOKS:

1. The Security Operations Center: Building, Operating, and Maintaining Your SOC, David Nathans
2. Network Security Through Data Analysis, Michael Collins
3. Blue Team Handbook: Incident Response Edition, Don Murdoch

REFERENCES:

1. Applied Network Security Monitoring: Collection, Detection, and Analysis, Chris Sanders and Jason Smith
2. The Practice of Network Security Monitoring: Understanding Incident Detection and Response, Richard Bejtlich

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CY402) WEB AND DATABASE SECURITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand an overview of web and database security
- To gain knowledge about privacy preserving techniques for web
- To learn about security re-engineering
- To be familiar with future trends in databases
- To explore privacy protection and security in Data publishing

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand web architecture, security, common attacks, session management, HTTPS, SSL/TLS, and countermeasures

CO-2: Identify application security, cyber-attacks, vulnerabilities, testing methods, and secure software development practices

CO-3: Examine database security, watermarking, trustworthy records, damage recovery, and Hippocratic database concepts and trends

CO-4: Identify the vulnerabilities in databases and enforce the Security and Privacy policies

CO-5: Interpret various techniques in database privacy

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	3	2	1	2	1	1	3	2	1	1	3	3	3	3
CO-2	2	2	3	2	2	3	1	3	1	1	2	3	2	3	3
CO-3	1	2	2	1	2	2	1	2	1	2	1	3	3	2	3
CO-4	3	3	3	2	3	3	1	3	1	1	1	2	3	3	3
CO-5	2	2	2	2	2	2	1	3	1	3	1	3	3	1	2

UNIT-I:

The Web Security: Introduction, Importance, Security architecture of World Wide Web, Security Architecture of Web Servers, and Web Clients Web Application Security – Cross Site Scripting Attacks, Cross Site Request Forgery, SQL Injection Attacks Content Security Policies (CSP) in web Session Management and User Authentication, Session Integrity Https, SSL/TLS

Side-Channel Attacks: Various side channels, Timing Analysis, Power analysis attack, Simple and differential power analysis, Counter measures.

UNIT-II:

Application Security: Introduction, Hexad, types of cyber attacks, Introduction to software development vulnerabilities, code analyzers – Static and dynamic analyzers,

Security Testing: Principles and concepts, PT workflows and examples, blind tests, synthetic transactions, interface testing and fuzzing, SDLC phases and security mandates.

UNIT-III:

Database Security: Introduction includes threats, vulnerabilities and breaches, Basics of database design, DB security – concepts, approaches and challenges, types of access controls, Oracle VPD, Discretionary and Mandatory access control – Principles, applications and poly-instantiation, Database inference problem, types of inference attacks, distributed database, security levels.

UNIT-IV:

Security Re-engineering for Databases: Concepts and Techniques, Database Watermarking for Copyright Protection, Trustworthy Records Retention, Damage Quarantine and Recovery in Data Processing Systems, Hippocratic Databases: Current Capabilities and Future Trends.

UNIT-V:

Privacy in Database Publishing: A Bayesian Perspective, Privacy-enhanced Location-based Access Control, Efficiently Enforcing the Security and Privacy Policies in a Mobile Environment.

TEXT BOOKS:

1. Web Security, Privacy and Commerce, Simson G. Arfinkel, Gene Spafford, O' Reilly, 2002
2. Handbook on Database Security Applications and Trends, Michael Gertz, Sushil Jajodia, Springer, 2008

REFERENCES:

1. Database and Applications Security, Bhavani Thuraisingham, Integrating Information Security and Data Management, Auerbach Publications, 2005
2. Web Security & Commerce, Simson Garfinkel, Gene Spafford, Editor Debby Russell, O'Reilly, 1997
3. Database and Applications Security Integrating Information Security and Data Management, Bhavani Thuraisingham, 2005, Auerbach Publications
4. Cryptography and Network Security: Principles and Practice, William Stallings, 7th Edition, Pearson, 2017
5. Database Security, Alfred Basta and Melissa Zgola, Cengage Learning, 2011

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1AM405) COGNITIVE ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To identify the basics of cognitive science engineering with focus on mind organization, brains
- To analyze human intelligence and artificial intelligence
- To understand basics of neuroscience and linguistics
- To analyze various levels of cognitive functioning and its applications
- To relate the concepts of cognitive field in robotics and real time applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Introduce role of computers in cognitive science and neuroscience applications

CO-2: Illustrate various information processing models and cognitive architecture

CO-3: Apply various computation tools for cognitive functioning

CO-4: Analyze cognitive science applications in image processing linguistic applications

CO-5: Demonstrate the use of cognitive models and patterns in distributed environment, and real time applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	2	1	1	1	2	1	1	1	2	2	2
CO-2	2	3	2	2	3	1	1	2	2	2	1	1	3	3	2
CO-3	2	3	3	2	2	1	2	1	2	2	2	1	3	3	3
CO-4	3	3	3	2	2	2	1	2	3	3	2	2	3	3	2
CO-5	2	2	2	3	2	1	2	1	2	2	3	2	2	2	3

UNIT-I:

Introduction to Cognitive Science: The Cognitive view –Some Fundamental Concepts – Computers in Cognitive Science – Applied Cognitive Science – The Interdisciplinary Nature of Cognitive Science, The Architecture of the Mind. Cognitive Processes, Computational Neuroscience - The Organization of the mind.

UNIT-II:

Higher-Level Cognition and Challenges: Dynamical systems and situated cognition- Challenges – Emotions and Consciousness – Computation of Cognitive Functioning in machines: Robotics, Human-Robotics Interaction.

UNIT-III:

Cognitive Science for Vision/Image Processing: Perception and sensing: visual cognition, cognitive mechanisms of vision, Feature Extraction from Images, Information Processing in Perception and Visual Behavior, Cognitive function measurement tools and software.

UNIT-IV:

Language Acquisition and Semantics: Milestones in acquisition, Theoretical perspectives, Semantics and Cognitive Science, Meaning and entailment, Sense, Problems in possible worlds Semantics, Cognitive and computational models of semantic Processing.

UNIT-V:

Distributed Cognitive Learning Using Fuzzy Cognitive Maps: Axelrod's cognitive maps, kosko's model, adaptive FCM, Pal and Konar's FCM model-examples.

TEXT BOOKS:

1. Cognitive Science: An Introduction, Neil Stillings, Steven E. Weisler, Christopher H. Chase and Mark H. Feinstein, 2nd Edition, 1995
2. Cognitive Science: An Introduction to the Science of the Mind José Luis Bermúdez, Cambridge University Press, 2010
3. Cognitive Engineering: A Distributed Approach to Machine Intelligence (Advanced Information and Knowledge Processing), Amit Konar, Springer, 2005

REFERENCES:

1. Computational Vision: Information Processing in Perception and Visual Behavior, Hanspeter A. Mallot, Translated by John S. Allen
2. How the Mind Works, Steven Pinker, 2009
3. Cognitive Science: An Interdisciplinary Approach, Carolyn Panzer Sobel and Paul Li, 2013
4. Mind: Introduction to Cognitive Science, Paul Thagard, 2nd Edition, MIT Press, 2004
Wilson Robert A., & Keil Frank C. (Eds.)
5. The MIT Encyclopedia of the Cognitive Sciences (MITECS), MIT Press, 2001

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CS404) AUGMENTED REALITY AND VIRTUAL REALITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Programming and Data Structures

COURSE OBJECTIVES:

- To understand the basic concepts of Virtual and Augmented Reality
- To identify the differences in AR/VR concepts and technologies
- To understand the fundamental concepts relating to Virtual Reality such as presence, immersion, and engagement
- To evaluate usability of AR/VR applications and critique their use of AR/VR capabilities
- To design and prototype effective AR/VR applications using UNITY platform for business, industry, non-profit and government organizations

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Introduce role of augmented and virtual reality in interactive application development

CO-2: Apply Geometric Modeling Techniques

CO-3: Implement virtual environment with respect to user experience

CO-4: Design Virtual/Augmented Reality Applications

CO-5: Acquire fundamental and hands-on skills needed to develop AR/VR applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	1	1	2	2	1	-	2	2	-	-	-
CO-2	1	1	1	3	1	2	2	1	2	1	1	2	-	-	-
CO-3	1	1	2	-	3	1	1	1	1	2	1	2	-	-	-
CO-4	-	2	-	-	-	2	-	2	1	1	3	1	1	-	-
CO-5	-	-	2	-	3	-	-	-	-	-	-	-	1	1	-

UNIT-I:

Introduction to Augmented Reality (AR): Definition and Scope A Brief History of Augmented Reality Displays (Multimodal Displays, Spatial Display Model, and Visual Displays) Strong vs Weak AR Applications of AR Challenges in AR

UNIT-II:

Introduction to Virtual Reality (VR): Definition and Scope Types of VR Characteristics of VR Basic VR environments Limitations of VR environments Immersion Vs Presence Key hardware requirements for VR

UNIT-III:

Interaction Design for AR/VR Environments: Interaction design process Identifying user needs AR/VR design considerations Typical AR/VR Interface Metaphors Affordances in AR/VR Human Information Processing Design for Perception and cognition User experience (UX) guidelines for AR/VR UX challenges for AR/VR Prototyping for AR/VR Evaluation of the developed AR/VR prototype

UNIT-IV:

Unity Overview: Windows, Interface, Navigation, Terminology, Game Objects, Hierarchy, Parenting Objects Asset Store, Importing Plug-ins Creating a Terrain, Materials, Colors, Transparency Introduction to MonoBehaviour: Awake, Start, Update

UNIT-V:

Introduction to Vuforia and Physics in UNITY

Vuforia Overview: Interface, Navigation, Terminology, Image Targeting, Custom Images Overview of Physics in Unity

Introduction to Scripting: Terminology, Creating Objects, Accessing Components, Debugging, Lists, Loops

TEXT BOOKS:

1. Understanding virtual reality, Sherman W. R., & Craig A. B., Morgan Kaufman, 2003
2. Augmented Reality: Principles and Practice, Schmalstieg D., & Hollerer T., Addison-Wesley Professional, 2016

REFERENCES:

1. A survey of augmented reality. Presence: Teleoperators & Virtual Environments, Azuma R. T., 1997, 6(4), 355–385
2. Recent Advances in Augmented Reality, IEEE Computer Graphics and Applications, Azuma R., Baillot Y., Behringer R., Feiner S., Julier S., & MacIntyre B., 2001, 21(6), 34-47
3. A Cost-Effective Interactive 3D Virtual Reality System Applied To Military Live Firing Training, Virtual Reality, Bhagat K. K., Liou, W. K., & Chang, C. Y, 2016, 20(2), 127-140. doi:10.1007/s10055-016-0284-x
4. Augmented Reality Technologies, Systems And Applications, Multimedia Tools and Applications, Carmigniani J., Furht B., Anisetti M., Ceravolo P., Damiani E., & Ivkovic M., 2011, 51(1), 341-377
5. Human Augmentation: Past, Present and Future. International Journal Of Human Computer Studies, Raisamo R., Rakkolainen I., Majaranta P., Salminen K., Rantala J., & Farooq, 2019, 131, 131-143. doi: <https://doi.org/10.1016/j.ijhcs.2019.05.008>

ONLINE RESOURCES:

1. Virtual Reality Engineering, Manivannan M., 2018, IIT Madras, <https://nptel.ac.in/courses/121106013>
2. Industry 4.0: Augmented Reality and Virtual Reality, Misra S., 2019, IIT Kharagpur, <https://www.youtube.com/watch?v=zLMgdYI82IE>
3. Augmented Reality - Fundamentals and Development, NPTEL Special Lecture Series, Dube A., 2020, <https://www.youtube.com/watch?v=MGuSTAqIZ9Q>
4. <http://cambum.net/course-2.htm>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1IT402) SOFTWARE PROJECT MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Software Engineering

COURSE OBJECTIVES:

- To identify and discuss the conventional and contemporary software project management principles
- To assess and plan project schedule and assign resources
- To apply an appropriate project development methodology among various alternating processes
- To identify project risks, understand the responsibilities, monitor and track project deadlines and the capability to work in a team environment

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Describe the conventional s/w management and explain how to improve s/w economics

CO-2: Identify and discuss the key phases of project management and the key skills associated with each

CO-3: Relate an appropriate project management approach through an evaluation of context and project scope and knowledge of agile and traditional and Global project management approaches, risk and quality management

CO-4: Apply the knowledge of the key project management skills, such as product and work break-down structure, schedule; governance, progress reporting and People Focused Process Models

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	2	1	1	2	1	2	-	-	-	1	1	1	3
CO-2	2	2	3	2	2	3	1	3	-	-	-	2	3	-	3
CO-3	2	2	2	3	3	1	2	2	-	-	-	2	-	3	-
CO-4	2	2	3	2	2	2	1	3	-	-	-	2	3	-	-
CO-5	2	2	3	2	2	2	1	1	-	-	-	2	1	1	3

UNIT-I:

Conventional Software Management: The waterfall model, Conventional software Management performance. Overview of Project Planning – Stepwise Project Planning.

Improving Software Economics: Reducing Software product size, improving software processes, improving team effectiveness, improving automation, Achieving required quality, peer inspections.

UNIT-II:

The Old Way and the New Way: The principles of conventional software Engineering, Principles of modern software management, Transitioning to an iterative process.

Life Cycle Phases: Engineering and Production stages, Inception, Elaboration, Construction, Transition phases.

Artifacts of the process: The artifact sets, Management artifacts, Engineering artifacts, Pragmatic artifacts.

Workflows of the Process: Software process workflows, Iteration workflows.

UNIT-III:

Checkpoints of the Process: Major milestones, Minor Milestones, Periodic status assessments.

Iterative Process Planning: Work breakdown structures, planning guidelines, cost and schedule estimating, Iteration planning process, Pragmatic planning.

UNIT-IV:

Project Organizations and Responsibilities: Line-of-Business Organizations, Understanding Behavior - Organizational Behavior, Project Organizations, Evolution of Organizations.

Process Automation: Automation building blocks, The Project Environment.

UNIT-V:

Project Control and Process Instrumentation: The seven core Metrics, Management indicators, quality indicators, life cycle expectations, pragmatic Software Metrics, Metrics automation

Future Software Project Management: Modern Project Profiles, Next generation Software economics, modern process transitions.

Emerging Trends in Project Management. Globalization Issues in Project management, People Focused Process Models.

TEXT BOOKS:

1. Software Project Management, Walker Royce, Pearson Education, 2005
2. Managing and global Software Projects, Ramesh Gopalaswamy, Tata McGraw-Hill, 2003

REFERENCES:

1. Software Project Management, Bob Hughes and Mike Cotterell, Tata McGraw-Hill Edition
2. Software Project Management, Joel Henry, Pearson Education
3. Software Project Management in Practice, Pankaj Jalote, Pearson Education, 2005

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B.Tech. VIII Semester

(22PE1AM402) REINFORCEMENT LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Machine Learning

COURSE OBJECTIVES:

- To understand multi-armed bandits
- To introduce finite Markov decision processes
- To understand dynamic programming and evaluation
- To understand Monte Carlo methods and decision process
- To analyze temporal difference learning and policy control

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Summarize the elements of Reinforcement Learning and apply K-Armed bandit problem for different situations

CO-2: Apply finite Markov decision process and model a control task in the Markov decision process framework

CO-3: Apply dynamic programming for policy evaluation and improvement

CO-4: Outline the Monte Carlo methods and sampling techniques

CO-5: Apply temporal difference for prediction models

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	1	2	-	-	-	-	-	2	2	2	2
CO-2	1	2	2	1	-	-	-	-	-	-	-	2	1	2	2
CO-3	2	2	3	-	2	1	-	-	-	-	-	2	2	1	2
CO-4	2	1	3	1	-	-	-	-	-	-	-	2	3	3	3
CO-5	2	2	3	-	1	1	-	-	-	-	-	2	3	3	3

UNIT-I:

Introduction to Reinforcement Learning: Reinforcement Learning, Elements of Reinforcement Learning, Limitations and Scope.

Multi-Armed Bandits: A K-Armed Bandit Problem, Action-Value Methods, 10-Armed Testbed, Incremental Implementation, Tracking a Nonstationary Problem, Optimistic Initial Values, Upper Confidence Bound Action Selection, Gradient Bandit Algorithm, Associative Search.

UNIT-II:

Finite Markov Decision Processes: Agent-Environment Interface, Goals and Rewards, Returns and Episodes, Unified Notation for Episodic and Continuing Tasks, Policies and Value Functions, Optimal Policies and Optimal Value Functions, Optimality and Approximation.

UNIT-III:

Dynamic Programming: Policy Evaluation, Policy Improvement, Policy Iteration, Value Iteration, Asynchronous Dynamic Programming, Generalized Policy Iteration, Efficiency of Dynamic Programming.

UNIT-IV:

Monte Carlo Methods: Monte Carlo Prediction, Monte Carlo Estimation of Action Values, Monte Carlo Control, Monte Carlo Control Without Exploring Starts, Off-Policy Prediction via Importance Sampling, Incremental Implementation, Off-Policy Monte Carlo Control, Discounting Aware Importance Sampling, Per- Decision Importance Sampling.

UNIT-V:

Temporal Difference Learning: Temporal Difference (TD) Prediction, Advantages of TD Prediction Models, Optimality of TD(0), Sarsa On-Policy TD Control, Q-Learning Off-Policy TD Control, Expected Sarsa, Maximization Bias and Double Learning, Games, Afterstates and Other Special Cases.

TEXT BOOK:

1. Reinforcement Learning: An Introduction, Richard S. Sutton, Andrew G. Barto, 2nd Edition, MIT Press

REFERENCE:

1. Reinforcement Learning, Phil Winder, 1st Edition, O'Reilly

ONLINE RESOURCES:

1. <https://www.microsoft.com/en-us/research/event/reinforcement-learning-day-2019/agenda/>
2. <https://deepmind.google/>
3. <https://web.stanford.edu/class/cs234/index.html>
4. <https://web.mit.edu/dimitrib/www/RLbook.html>
5. <https://www.codementor.io/reinforcement-learning-experts>

B.Tech. VIII Semester

(22PE1CY403) SECURITY IN INTERNET OF THINGS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand IoT basic concepts and security related issues
- To analyze security concerns in IoT, focusing on authentication, encryption, lightweight cryptography, and challenges across multiple IoT layers
- To examine security concerns, challenges, and solutions in IoT environments
- To investigate various security issues in social IoT, Healthcare IoT and Data
- To understand cloud services for IoT security, focusing on offerings, security controls, and IoT cloud architecture

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze various security issues in Internet of Things

CO-2: Investigate study of security architecture and node authentication in Internet of Things

CO-3: Implement enabling Technologies to meet security requirements in Internet of Things

CO-4: Design and apply security models in Social IoT, Healthcare IoT and Data Security

CO-5: Evaluate and design IoT cloud security solutions, assessing offerings and architecture for cloud-enabled IoT computing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	3	2	1	2	3	1	3	2	1	1	3	3	1	3
CO-2	2	2	3	2	3	3	1	3	1	1	2	3	3	3	3
CO-3	1	2	2	1	2	3	1	3	3	2	1	3	3	1	3
CO-4	3	3	3	3	3	3	1	3	1	1	1	3	2	1	3
CO-5	2	2	2	2	2	2	1	3	1	1	1	3	2	3	3

UNIT-I:

Introduction to Internet of Things (IoT): IoT definition, Characteristics of IoT, Physical Design of IoT: Things in IoT, IoT Protocols, Logical Design of IoT: IoT Functional Blocks, IoT Communication Models, IoT Communication APIs.

UNIT-II:

Security Concerns in IoT: Security Goals and Challenges in IoT environment, Identity based Authentication, Encryption and Digital Signature, IP Connectivity, Lightweight Cryptography, Existing Security Schemes for IoT, Security Requirements in IoT Architecture, Security challenges across multiple IoT layers, including network, service, application, and cross-layer, Threats caused in Maintenance of IoT.

UNIT-III:

Security for Enabling Technologies: Security in Identification and Tracking Technologies, Security in integration of Wireless sensor Networks and RFID, Security in Communications, Security Protocols and Privacy Issues into 6LoWPAN stack, Security in Service Management.

UNIT-IV:

Security in Social IoT and Healthcare: Existing Security Scheme for IOT, Security Concerns in Social IoT, Confidentiality and Security for IoT based Healthcare. IoT Data Security.

UNIT-V:

Cloud for IoT Security: Cloud services and IoT, Offerings related to IoT from cloud service providers, Cloud IoT security controls, An enterprise IoT cloud security architecture, New directions in cloud enabled IoT computing

TEXT BOOKS:

1. Internet of Things, A Hands-on Approach, Vijay Madisetti, Arshdeep Bahga, University Press
2. Securing the Internet of Things, Shancang Li, Li Da Xu, Syngress Publications, An Imprint of Elsevier, 2017
3. Security and Privacy in Internet of Things (IoT)-Models, Algorithms and Implementations, Fei Hu, CRC Press, Taylor & Francis Group, 2016

REFERENCES:

1. Internet of Things for Architects: Architecting IoT Solutions by Implementing Sensors, Communication Infrastructure, Edge Computing, Analytics, and Security, Perry Lea, Packt Publishing, 2018
2. IoT Security Issues, Alasdair Gilchrist, 1st Edition, DEG Press, 2017
3. IoT Security Paradigms and Applications – Research and Practices, Bharat Bhushan, Narayan C. Debnath, Sudhir Kumar Sharma, CRC Press, 2020
4. IoT Security and Privacy Paradigm, Dac-Nhuong Le, Souvik Pal, Vicente Garcia Diaz, CRC Press, 2020

B.Tech. VIII Semester

(22PE1IN303) ADHOC AND SENSOR NETWORKS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To learn about the issues and challenges of wireless Adhoc networks
- To understand the working of MAC and Routing Protocols for Adhoc and sensor networks
- To learn about the Transport Layer protocols and their QoS for Adhoc and sensor networks
- To understand various security issues in Adhoc and sensor networks and the corresponding solutions
- To understand the various energy management schemes in Adhoc and sensor networks

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand basic concepts and challenges related to Adhoc sensor networks

CO-2: Analyze architecture and issues in designing protocols for Adhoc and sensor networks

CO-3: Identify and analyze various quality services and security issues of Adhoc and sensor networks

CO-4: Relate the applications and energy management schemes for Ad hoc sensor networks

CO-5: Implement different protocols in real time applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	-	-	-	-	-	-	-	-	-	-	1	2	1
CO-2	-	-	-	1	-	-	-	-	-	1	-	-	2	1	-
CO-3	-	-	-	1	-	-	1	1	-	-	-	-	-	1	-
CO-4	-	1	-	-	1	1	-	-	-	-	-	-	1	1	2
CO-5	2	2	2	-	1	-	-	1	1	1	1	2	1	1	3

UNIT-I:

Introduction to AdHoc Wireless Networks: Introduction, Issues in AdHoc Wireless Networks. Ad Hoc Wireless Internet. Summary. Problems. Bibliography.

MAC Protocols for AdHoc Wireless Networks: Introduction, Issues in Designing a MAC Protocol for Ad Hoc Wireless Networks. Design Goals of a MAC Protocol for Ad Hoc Wireless Networks. Classifications of MAC Protocols. Contention-Based Protocols. Contention-Based Protocols with Reservation Mechanisms. Contention-Based MAC Protocols with Scheduling Mechanisms. MAC Protocols That Use Directional Antennas. Other MAC Protocols

UNIT-II:

Routing Protocols for AdHoc Wireless Networks: Introduction. Issues in Designing a Routing Protocol for Ad Hoc Wireless Networks. Classifications of Routing Protocols. Table-Driven Routing Protocols. On-Demand Routing Protocols. Hybrid Routing Protocols. Routing Protocols with Efficient Flooding Mechanisms. Hierarchical Routing Protocols. Power-Aware Routing Protocols.

UNIT-III:

Multi-cast routing in AdHoc Wireless Networks: Introduction. Issues in Designing a Multicast Routing Protocol. Operation of Multicast Routing Protocols. An Architecture Reference Model for Multicast Routing Protocols. Classifications of Multicast Routing Protocols. Tree-Based Multicast Routing Protocols. Mesh-Based Multicast Routing Protocols. Summary of Tree-and Mesh-Based Protocols. Energy-Efficient Multicasting. Multicasting with Quality-of-Service Guarantees. Application-Dependent Multicast Routing.

UNIT-IV:

Transport Layer and Security Protocols for AdHoc Wireless Networks: Introduction. Issues in Designing a Transport Layer Protocol for Ad Hoc Wireless Networks. Design Goals of a Transport Layer Protocol for Ad Hoc Wireless Networks. Classification of Transport Layer Solutions. TCP Over Ad Hoc Wireless Networks. Other Transport Layer Protocols for Ad Hoc Wireless Networks. Security in Ad Hoc Wireless Networks. Network Security Requirements. Issues and Challenges in Security Provisioning. Network Security Attacks. Key Management. Secure Routing in Ad Hoc Wireless Networks

UNIT-V:

Quality of Service and Energy Management in AdHoc Wireless Networks: Classifications of QoS Solutions. MAC Layer Solutions. Network Layer Solutions. QoS Frameworks for Ad Hoc Wireless Networks, Classification of Energy Management Schemes. Battery Management Schemes. Transmission Power Management Schemes. System Power Management Schemes.

TEXT BOOKS:

1. AdHoc Wireless Networks – Architectures and Protocols, C. Siva Ram Murthy and B. S. Manoj, Pearson Education, 2006
2. Protocols and Architectures for Wireless Sensor Networks, Holger Karl, Andreas Willing, John Wiley & Sons, 2005

REFERENCES:

1. AdHoc Mobile Wireless Networks, Subir Kumar Sarkar, T. G. Basavaraju, C. Puttamadappa, Auerbach Publications, 2008
2. AdHoc and Sensor Networks: Theory and Applications, Carlos De Moraes Cordeiro, Dharma Prakash Agrawal, 2nd Edition, World Scientific, 2011
3. Fundamentals of Wireless Sensor Networks Theory and Practice, Waltenegus Dargie, Christian Poellabauer, John Wiley and Sons, 2010
4. Wireless AdHoc and Sensor Networks: Theory and Applications, Xiang-Yang Li, Cambridge University Press, 2008

B.Tech. VIII Semester

(22PE1DS406) QUANTUM COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basics of quantum computation and quantum mechanics
- To apply different models of quantum computation
- To illustrate the quantum models using machine learning algorithms
- To explore various Quantum algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explore the quantum computation environment

CO-2: Identify the implications of quantum computation in the field of mathematics and associated fields

CO-3: Demonstrate building blocks and programming methodologies for quantum computing

CO-4: Apply quantum computation algorithms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	3	2	-	-	2	2	2	3	3	1	3
CO-2	2	2	3	2	3	3	-	-	2	2	2	3	3	2	3
CO-3	2	2	3	3	3	3	-	-	3	2	3	2	3	2	1
CO-4	3	2	2	3	2	2	-	-	3	2	2	3	3	2	3

UNIT-I:

Introduction and Background: Overview, The Circuit Model of Computation, A Linear Algebra Formulation of the Circuit Model, Reversible Computation. A Preview of Quantum Physics.

Linear Algebra and The Dirac Notation: The Dirac Notation and Hilbert, Dual Vectors, Operators, The Spectral Theorem, Functions of Operators

UNIT-II:

Qubits and The Framework of Quantum Mechanics: The State of a Quantum System, Time-Evolution of a Closed System, Composite Systems.

A Quantum Model of Computation: The Quantum Circuit Model, Quantum Gates, 1-Qubit Gates, Controlled-U Gates, Universal Sets of Quantum Gates, Efficiency of Approximating Unitary Transformations.

UNIT-III:

Superdense Coding and Quantum Teleportation: Superdense Coding, Quantum Teleportation, An Application of Quantum Teleportation.

Quantum Error Correction: Error Models for Quantum Computing, Encoding, Error Recovery. Architecture: Bits and Qubits, Classical Gates, Reversible Gates, Quantum Gates.

UNIT-IV:

Programming Languages: Programming in a Quantum World, Quantum Assembly Programming, Toward Higher-Level Quantum Programming.

Theoretical Computer Science: Deterministic and Nondeterministic Computations, Probabilistic Computations, Quantum Computations.

UNIT-V:

Quantum Algorithms: Deutsch's Algorithm, The Deutsch–Jozsa Algorithm, Simon's Periodicity Algorithm, Grover's Search Algorithm, Shor's Factoring Algorithm

TEXT BOOKS:

1. An Introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, Michele Mosca, Oxford University Press
2. Quantum Computing for Computer Scientists, Noson S. Yanofsky, Michael A. Mannucci, Cambridge University Press

REFERENCES:

1. Quantum Computing, Ronald De Wolf, QuSoft, CWI and University of Amsterdam
2. Quantum Computing Explained, David McMahon, Wiley
3. Quantum Computing Fundamentals, William (Chuck) Easttom

B.Tech. VIII Semester

(22PE1AM407) ROBOTIC PROCESS AUTOMATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To explore the knowledge of basic concepts of Robotic Process Automation
- To outline the RPA design & development strategies
- To apply design methodologies adopted by UiPath
- To develop flow diagrams using RPA for real-world application
- To apply programming robots for logging and exception handling

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand Robotic Process Automation concepts

CO-2: Adopt UiPath programming techniques to deploy robot configurations

CO-3: Explore various data extraction techniques and perform integrations with various popular applications

CO-4: Design and develop a programmed robot that includes logging and exception handling

CO-5: Deploy and control Bots with UiPath orchestrator

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	1	3	3	2	2	2	1	1	2	2	2	3	3
CO-2	2	2	2	1	3	1	1	2	1	1	1	1	2	2	2
CO-3	1	2	2	3	2	2	1	3	1	2	1	2	2	2	2
CO-4	2	2	3	3	2	2	1	2	1	2	1	2	2	1	2
CO-5	1	2	1	1	3	1	1	2	1	2	1	1	2	2	2

UNIT-I:

Introduction: What is Robotic Process Automation (RPA), Scope & techniques of Automation, Benefits of RPA, Components of RPA, RPA Platforms, UiPath Studio, Installation of UiPath Studio, Learning UiPath Studio.

UNIT-II:

Sequence, Flowchart & Control Flow: Sequencing the Workflow, Activities, Flowchart, Control Flow for Decision making. Data Manipulation: Variables, Collection, Arguments, Data Table, Clipboard management, File operations with step-by-step example

Controls: Finding the control, waiting for a control, Act on a control, UiExplorer, Handling Events Recording and advanced UI Interaction Basic recording, Desktop recording, Web recording, Citrix, Screen Scraping, when to use OCR, Types of OCR available, how to use OCR Avoiding typical failure points.

UNIT-III:

Plugins and Extensions: Terminal plugin, Java plugin, Java plugin with UiPath Studio, Citrix automation, Citrix environment, Mail plugin, PDF plugin, Web integration, Excel and Word plugins, Credential management Extensions Handling User Events and Assistant Bots: What are assistant bots, Monitoring system event triggers: Hotkey trigger, Mouse trigger, System trigger, Monitoring image and element triggers, Launching an assistant bot on a keyboard event.

UNIT-IV:

Exception Handling, Debugging, and Logging: Exception handling, Common exceptions and ways to handle them, Logging and taking screenshots Debugging techniques, setting break points, Slow step, Highlighting, Break, collecting crash dumps: Enabling crash dumps, Disabling crash dumps, Error reporting: Enterprise Edition customers, Community Edition users. Managing and Maintaining the Code: Project organization, picking an appropriate layout for each workflow, Breaking the process into smaller parts, Using exception handling, Making your workflow readable, Keeping it clean, Nesting workflows, Reusability of workflows, Templates, Commenting techniques, State Machine, When to use Flowcharts State Machines or Sequences.

UNIT-V:

Deploying and Maintaining the Bot: Publishing using publish utility, Overview of Orchestration Server, Using Orchestration Server to control bots, Using Orchestration Server to deploy bots, License management, Activating and uploading a license to Orchestrator, Publishing and managing updates, Packages, Managing packages.

TEXT BOOK:

1. Learning Robotic Process Automation, Alok Mani Tripathi, O'Reilly Publishing, 2018

REFERENCES:

1. Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant, Richard Murdoch, Amazon Asia-Pacific Holdings, 2018
2. Introduction to Robotic Process Automation: A Primer, Institute of Robotic Process Automation, Frank Casale, Rebecca Dilla, Heidi Jaynes, Lauren Livingston, Amazon Asia-Pacific Holdings, 2018

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B.Tech. VIII Semester

(22PE1CS405) SOCIAL MEDIA ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Graph Theory, Data Mining, Python/R programming

COURSE OBJECTIVES:

- To familiarize with the concept of social media
- To familiarize with the concept of social media analytics and understand its significance
- To enable to develop skills required for analyzing the effectiveness of social media
- To examine the ethical and legal implications of leveraging social media data
- To familiarize with different visualization techniques for Social media analytics

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the concept of social media analytics and its significance

CO-2: Analyze the effectiveness of social media

CO-3: Use different effective visualization techniques to represent social media analytics

CO-4: Use different social media analytics tools effectively and efficiently

CO-5: Acquire the fundamental perspectives and hands-on skills needed to work with social media data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	3	3	2	-	-	2	2	2	3	3	3	3
CO-2	3	2	3	2	3	3	-	-	2	2	2	3	3	3	3
CO-3	3	2	3	3	3	3	-	-	3	2	3	2	3	3	3
CO-4	3	2	2	3	2	2	-	-	3	2	2	3	3	2	2
CO-5	3	2	2	3	2	3	-	-	2	2	3	2	3	2	3

UNIT-I:

Social Media Analytics: An Overview: Core Characteristics of Social Media, Types of Social Media, Social media landscape, Need for Social Media Analytics (SMA), SMA in small & large organizations. Purpose of Social Media Analytics, Social Media vs. Traditional Business Analytics, Seven Layers of Social Media Analytics, Types of Social Media Analytics, Social Media Analytics Cycle, Challenges to Social Media Analytics, Social Media Analytics Tools

UNIT-II:

Social Network Structure, Measures & Visualization: Basics of Social Network Structure - Nodes, Edges & Tie Describing the Networks Measures - Degree Distribution, Density, Connectivity, Centralization, Tie Strength & Trust Network Visualization - Graph Layout,

Visualizing Network features, Scale Issues. Social Media Network Analytics - Common Network Terms, Common Social Media Network Types, Types of Networks, Common Network Terminologies, Network Analytics Tools.

UNIT-III:

Social Media Text, Action & Hyperlink Analytics: Social Media Text Analytics - Types of Social Media Text, Purpose of Text Analytics, Steps in Text Analytics, Social Media Text 8 Analysis Tools Social Media Action Analytics - What Is Actions Analytics? Common Social Media Actions, Actions Analytics Tools Social Media Hyperlink Analytics - Types of Hyperlinks, Types of Hyperlink Analytics, Hyperlink Analytics Tools

UNIT-IV:

Social Media Location & Search Engine Analytics: Location Analytics - Sources of Location Data, Categories of Location Analytics, Location Analytics and Privacy Concerns, Location Analytics Tools Search Engine Analytics - Types of Search Engines, Search Engine Analytics, Search Engine Analytics Tools

UNIT-V:

Social Information Filtering: Social Information Filtering - Social Sharing and filtering, Automated Recommendation systems, Traditional Vs social Recommendation Systems Understanding SocialMedia and Business Alignment, Social Media KPI, Formulating a Social Media Strategy, Managing Social Media Risks

TEXT BOOKS:

1. Seven Layers of Social Media Analytics - Mining Business Insights from Social Media Text, Actions, Networks, Hyperlinks, Apps, Search Engine, and Location Data, Gohar F. Khan, 2015
2. Mining the Social Web - Analyzing Data from Facebook, Twitter, LinkedIn, and Other Social Media Sites, Matthew A. Russell, O'Reilly
3. Social Network Data Analytics, Charu Aggarwal (Ed.), Springer, 2011

REFERENCES:

1. Analyzing the Social Web, Jennifer Golbeck, 1st Edition
2. Social Media Analytics, Techniques and Insights for Extracting Business Value Out of Social Media, Matthew Ganis, Avinash Kohirkar, IBM Press, 2015
3. Social Media Analytics Strategy - Using Data to Optimize Business Performance, Alex Gonçalves, APress
4. Social Media Data Mining and Analytics, Szabo G., G. Polatkan, O. Boykin & A. Chalkiopoulos, Wiley, 2019
5. Social Media Analytics: Effective Tools for Building, Interpreting and Using Metrics, Marshall Sponder, 1st Edition, McGraw-Hill, 2011

ONLINE RESOURCES:

1. <https://cse.iitkgp.ac.in/~pawang/courses/SC16.html>
2. https://onlinecourses.nptel.ac.in/noc20_cs78/preview